Anish Bhat

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Education

University of California, San Diego *Expected Graduation, June 2028*
B.S. Physics w/ Specialization in Computational Physics
- **Concentrations**: Applications of AI/ML in Quantum Computing and Robotics
- **Relevant Coursework:** Scientific Computing in Python, Linear Algebra, Multivariable Calculus, Differential Equations, Mechanics

Experience

Duarte Lab at UC San Diego *San Diego, CA*
Undergraduate Researcher *Jan 2026 – Present*
- Trained a Dynamic Graph Convolutional Neural Network (DGCNN) to classify particle-detector clusters from rare versus common physics processes at the Large Hadron Collider using point-cloud representations
- Optimized a Boosted Decision Tree baseline to benchmark learning performance, allowing a comparison between learned representations and hand-engineered physics features

Profluento *Austin, TX*
Software & Machine Learning Intern *Jul 2025 – Oct 2025*
- Deployed a speech-to-text API using TypeScript and OpenAI Whisper, enabling 10+ active users to convert spoken input into accurate text
- Optimized API performance to process and transcribe 500+ words efficiently, reducing manual transcription time by 35%
- Implemented backend architecture with error handling and caching mechanisms to ensure high availability and low latency in production use

Projects

Layer-Resolution of Sensitivity and Locality in Finite-Depth VQCs *Oct 2025 - Jan 2026*
2026 National Undergraduate Research Conference at Brown University
- Developed layer-resolved gradient diagnostics to quantify trainability in variational quantum circuits, revealing 6–8× stronger sensitivity for local vs. global observables across circuit depth
- Applied commutator-based sensitivity analysis to chart parameter influence through layers of finite-depth circuits, showing a near-doubling of global-to-local gradient ratios with depth considered
- Implemented quantum circuit simulations to compare observable locality effects, producing quantitative evidence correlating circuit structure to optimization behavior

Portable Nano Scanning Technology *Sep 2023 – Feb 2024*
Third place in Materials Science category at GARSEF 2024
- Engineered a compact and working model of the Scanning Tunneling Microscope in $75
- Modeled a motorized micro-positioning frame for 6 nm z-axis resolution using CAD and precision machining
- Used PWM signal control to implement feedback loops that automate height adjustment
- Implemented Fast Fourier Transform (FFT) algorithms to convert tunneling current signals into frequency-domain data

Skills

Languages: Python, C, Java, SQL, MATLAB, R

Tools and Frameworks: PyTorch, XGBoost, NumPy, Pandas, Matplotlib, Hugging Face, CAD, Qiskit

Applications: Deep Learning, Generative AI, Computer Vision, Natural Language Processing (NLP), Engineering Design, Circuit Design (PCB), Fourier Transform, Quantum Information and Variational Circuits